

RDL-like sweep through inversion (msinv hull simulator)

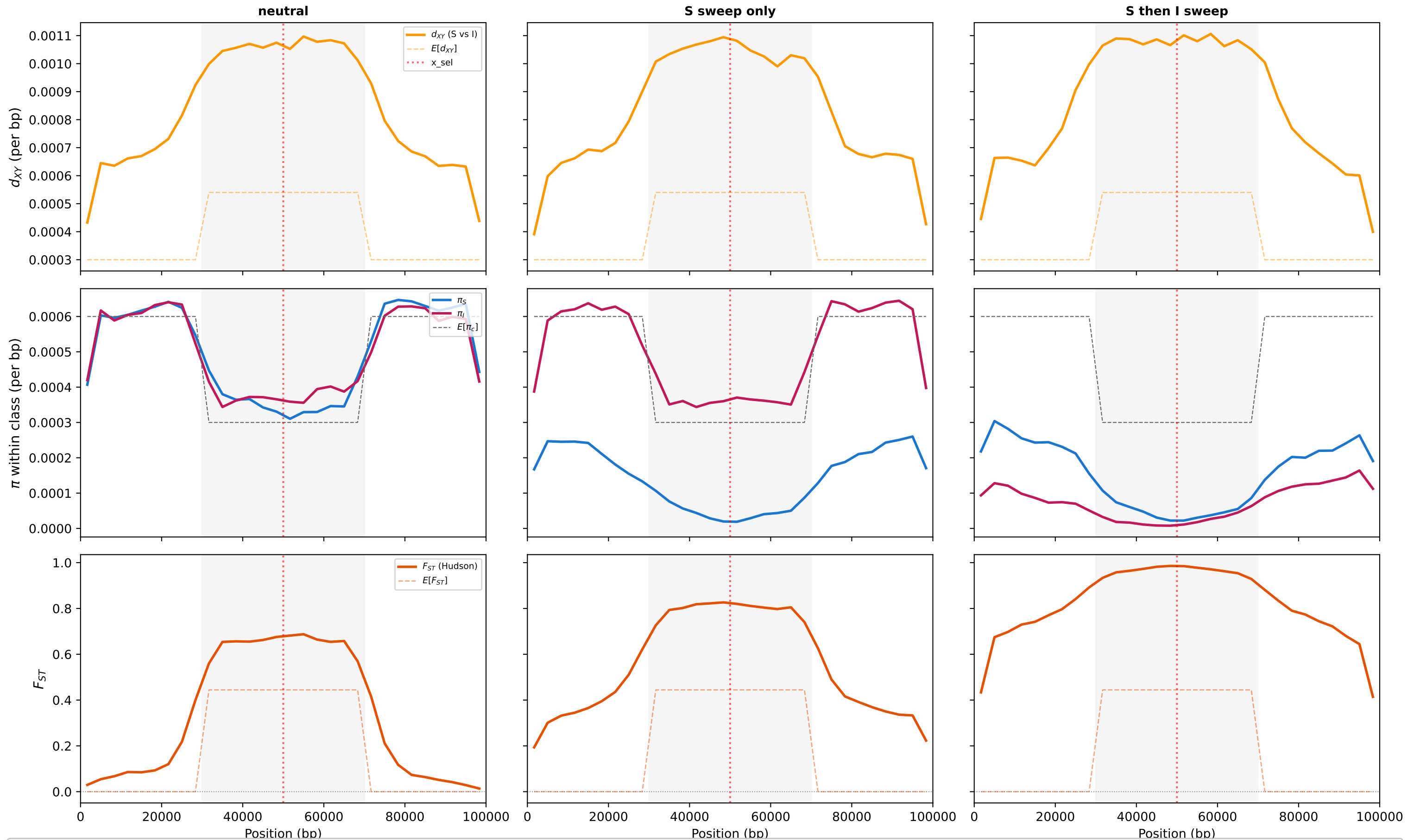


Figure. RDL (dieltrin resistance) sweep-through-inversion, modelled after Grau-Bové et al. (2020 MBE). Hitchhiking mode ($s=0.05$) with spatial decay $P(\text{linked}) = \exp(-r |x - x_{sel}| t_{dur})$. Three scenarios for $n_S=8 + n_I=8$ haplotypes. (Row 1) Cross-class d_{XY} . (Row 2) Within-class π_S , π_I . (Row 3) Hudson F_{ST} . Dashed lines: structured coalescent predictions (θ , $p \cdot \theta$, $\mu(t_{inv} + 2N_e)$, $F_{ST} = t_{inv}/(t_{inv} + 2N_e)$). Left: Neutral baseline — π dips to $p \cdot \theta$ inside inv (isolated classes). Centre: S sweep only ($t=700$ gen) — π_S valley around x_{sel} , π_I unaffected. Right: S then I sweep ($t_S=700$, $t_I=300$ gen) — both collapse. Parameters: $N_e=50,000$, $p_{inv}=0.5$, $t_{inv}=80,000$ gen, $s=0.05$, $\gamma=1e-9$, $\mu=3e-09$, $r=1e-08$, $L=100$ kb, 100 reps. Command: `pixi run -e all python examples/empirical_rdl_sweep.py`